Parameter Estimation for Linear Regression Type Model Structures – Effect of Model Structure Error

(具有线性回归类型模型结构的参数估计 – 模型结构误差的影响)

LSE under Effect of No Model Structure Error

- **No Model Structure Error** (没有模型结构误差)

$$Y(t) = \Phi(t)\theta^o$$

- **Necessary Condition of Model Parameters Accurately Recovered** (模型参数准确复现的必要条件)

$$\hat{\theta}(t) = P(t)\Phi(t)^T \overline{W}(t) Y(t)$$

$$P(t) = \left[\Phi(t)^T \overline{W}(t)\Phi(t)\right]^{-1} \in R^{p\times p}$$

$\Phi(t)^T \overline{W}(t)\Phi(t)$ **is nonsingular** (非奇异) $\Leftrightarrow$ $\Phi(t)$ **has a full column rank** p (是列满秩p)

$$\Downarrow$$

$$\hat{\theta}(t) = \theta^o$$

LSE under Effect of No Model Structure Error

- **Minimum Conditions for Experimental Design and Data Processing**
  (实验设计和数据处理的最小条件)

E1. ***The data set has to be large enough*** (数据集要足够大)

For single output model (m=1),the number of data sampling instances has to be no less than p(i.e., $t \geq p$) (对单一输出模型，瞬时采样获得的数据个数应不小于p)

$\Phi(t) \in R^{t \times p}$ has a full column rank p (**$\Phi(t)$是列p满秩**)

E2. ***The data set has to be rich enough*** (数据集要足够充分)

Remove segments of experimental data in which columns of $\Phi(t)$ are strongly correlated with each other (剔除使 **$\Phi(t)$** 具有强相关列的部分实验数据)

All columns of $\Phi(t)$ have to be linearly independent for it to have a rank of p (**$\Phi(t)$** 所有列必须线性无关，是列p满秩)

LSE under Effect of Model Structure Error

- **Actual Process** (实际过程)

$$y(t) = \varphi(t)^T \theta^0 + e(t)$$

$e(t)$ represents the model error ($e(t)$表示模型误差)

- **LSE** (最小二乘估计) **(assume** (假设) $W(t, i) = I_p$ **)**

$$\hat{\theta}(t) = \theta^0 + P(t)\left[\sum_{i=1}^{t} \varphi(i)e(i)\right]$$

$$\hat{\theta}(t) = P(t)\Phi(t)^T \overline{W}(t)Y(t) = P(t)[\sum_{i=1}^{t} \varphi(i)W(t,i)y(i)]$$

- **Define an Expectation-Like Operation Symbol** (定义类似期望的操作符号)

$$\overline{E}[s(t)] = \lim_{N\to\infty} \frac{1}{N}\sum_{t=1}^{N} E[s(t)]$$

- **Covariance-Function-Like Quantity** (类似协方差函数的量)

$$R_s(\tau) := \overline{E}[s(t)s(t-\tau)^T]$$

LSE under Effect of Model Structure Error

- **LSE Estimation Error under Model Structure Error** (有模型结构误差下的最小二乘估计误差)

$$\tilde{\theta}(t) = \hat{\theta}(t) - \theta^o = \left(\frac{1}{t}\sum_{i=1}^{t}\varphi(i)\varphi(i)^T\right)^{-1}\left(\frac{1}{t}\sum_{i=1}^{t}\varphi(i)e(i)\right) \xrightarrow{t\to\infty} R_\varphi(0)^{-1}R_{\varphi e}(0)$$

with large data set (具有较大数据集)

where, $R_{\varphi e}(\tau) = \overline{E}[\varphi(t)e(t-\tau)^T],\ \forall\tau$

- **Practical Means to Improve Parameter Estimation Accuracy** (提高参数估计精度的实用方法)

E3. ***Un-correlate the regressor vector φ(i) and the model error vector e(i)*** (不相关的回归向量φ(*i*)和模型误差向量*e*(*i*))

Uncorrelated $\varphi(i)$ and $e(i)$ with $E[e(t)] = 0$ (φ(*i*)和*e*(*i*)不相关) → $R_{\varphi e}(0) = 0$ →

Persistent exciting (PE) condition (持续激励条件) → $R_\varphi(0)$ is nonsingular (非奇异) →

An unbiased estimate (i.e., $\theta(t) \to \theta^0$ as $t \to \infty$) is obtained (可获得无偏估计)

LSE under Effect of Model Structure Error

E4 . ***Maximize the "signal to noise ratio" (SNR)*** **(最大化信噪比)** $\tilde{\theta}(t) \xrightarrow{t\to\infty} R_{\varphi}(0)^{-1} R_{\varphi e}(0)$

✓ **Useful Signal and Noise (有用信号和噪声)**

$\varphi(t)^T \theta^0$: Useful signal (the output due to the parameters to be estimated) **(有用信号(由待估计参数引起的输出))**

$e(t)$: Noise **(噪声)**

✓ **A measure of SNR (信噪比的测量)**

$$\bar{E}\left[\left\|\varphi(t)^T \theta^0\right\|^2\right] = \theta^{0T} R_{\varphi}(0)\theta^0 \quad \geq \quad \lambda_{\min}[R_{\varphi}(0)] \| \theta^0 \|^2$$

$$\bar{E}\left[\left\|e(t)\right\|^2\right] = Tr\{R_e(0)\} \quad \leq \quad m\lambda_{\max}[R_e(0)]$$

$\lambda_{\min}[R_{\varphi}(0)] / \lambda_{\max}[R_e(0)]$ ⟵

✓ **Practical Ways to Maximize SNR (使信噪比最大的实用方法)**

(i) Using as large inputs as possible to increase $\varphi(t)$ while at the same time making sure that $e(t)$ is kept small by not significantly violating the assumptions of the chosen model structure **(在与所选择的模型结构假设没有显著冲突的情况下，采用尽可能大的输入来增大$\varphi(t)$，同时使$e(t)$保持尽可能小)**

(ii)Proper filtering input and output data to remove the effects of major disturbances **(对输入和输出数据的适当滤波来消除主要干扰的影响)**

Converging Rate of LSE

- **Significance of the Converging Rate (收敛率的重要性)**
  - *all practical experiments run within finite time periods* (实际实验都是在有限时间内进行的)
- **Covariance Matrix of the LSE Estimation Error (最小二乘估计误差的协方差矩阵)**

$$\operatorname{cov}\left(\tilde{\theta}(t)\right) = E\left\{\tilde{\theta}(t)\tilde{\theta}(t)^T\right\} = E\left\{P(t)\left[\sum_{i=1}^{t}\sum_{j=1}^{t}\varphi(i)e(i)e(j)^T\varphi(j)^T\right]P(t)^T\right\}$$

$$= P(t)\left[\sum_{i=1}^{t}\sum_{j=1}^{t}\varphi(i)\sigma^2\delta_{ij}\varphi(j)^T\right]P(t) = \sigma^2 P(t) = \frac{1}{t}\sigma^2[tP(t)] = \frac{1}{t}\sigma^2\left[\frac{1}{t}\sum_{i=1}^{t}\varphi(i)\varphi(i)^T\right]^{-1} \xrightarrow{as\ t\to\infty} \frac{1}{t}\sigma^2 R_\varphi(0)^{-1}$$

- **Converging Rate of the LSE Estimation (最小二乘估计的收敛率)**

  (i) The covariance matrix of the LSE estimation error converges to zero at a rate of 1/t when the PE condition is satisfied, (i.e., $R_\varphi(0)$ is nonsingular) **(当持续激励满足时，LSE估计误差的协方差矩阵以1/*t*的速率收敛于零)**

  (ii) The larger $R_\varphi(0)$ is or the higher the signal excitation level is, the more accurate the resulting LSE parameter estimates tend to be! **(越大的$R_\varphi(0)$或越高的激励值，最小二乘参数估计越精确)**

Ex1. Effect of Experimental Conditions on LSE

- **Linear Regression Model with Single Parameter** (单一参数的线性回归模型)

$\hat{y}(t,\theta)=\varphi(t)\hat{\theta}$ **with an actual process of** (实际过程): $y(t)=\varphi(t)\theta^0+e(t)$

$$\Phi(t)^T\Phi(t)=\sum_{i=1}^{i=t}\varphi(i)^2 \Rightarrow \operatorname{cov}\left(\tilde{\theta}(t)\right)=\frac{\sigma^2}{\sum_{i=1}^{i=t}\varphi(i)^2}$$

$$\Rightarrow \text{If } \sum_{i=1}^{i=t}\varphi(i)^2\to\infty \text{ as } t\to\infty \;\Rightarrow\; \operatorname{cov}\left(\tilde{\theta}(t)\right)\to 0 \;\Rightarrow\; \hat{\theta}(t) \text{ converges to its true value } \theta^o$$

(收敛于真值)

- **Observations** (观察)

(1) If $\varphi(i)\sim e^{-ai}$, $a>0$

$\Rightarrow$ Covariance goes to a constant and parameter estimation does not converge (协方差趋向于定值，参数估计不收敛)

(2) If $\varphi(i)\sim 1$,

$\Rightarrow$ Covariance goes to zeros at a rate of $1/t$ (协方差以$1/t$速率趋向于零)

(3) If $\varphi(i)\sim e^{ai}$, $a>0$

$\Rightarrow$ Covariance goes to at a exponential rate of e^{-2at} (协方差趋向于e^{-2at}的指数速率)

Ex1. Effect of Experimental Conditions on LSE

- **Factors Affecting the Converging Rate** (收敛速率的影响因素)
  - The richness of the related signals (evaluated by signal to noise ratio) (相关信号的丰富性(由信噪比来评估))

$$SNR=\sqrt{\frac{\frac{1}{t}\sum_{i=1}^{i=t}\left(\varphi(i)\theta^0\right)^2}{\sigma^2}}=\sqrt{\frac{\frac{1}{t}\sum_{i=1}^{i=t}\varphi(i)^2}{\sigma^2}}\theta^0 \qquad \Rightarrow \qquad \operatorname{cov}\left(\tilde{\theta}(t)\right)=\frac{\theta^0}{SNR^2}\frac{1}{t}$$

$$\operatorname{cov}\left(\tilde{\theta}(t)\right)=\frac{\sigma^2}{\sum_{i=1}^{i=t}\varphi(i)^2}$$

(i) The larger the SNR is, the smaller the covariance matrix, and the more consistent the resulting parameter estimates are (信噪比越大，协方差越小，参数估计结果越一致)

(ii) Maximize SNR by (使信噪比最大的方法)

- using as large inputs as possible to make φ(t) bigger (尽可能用大的输入量使φ(t)较大)
- making sure that the model structure assumptions are not violated to keep the model error $e(t)$ small (确保模型结构假设不冲突以保持较小的模型误差)

Ex1. Effect of Experimental Conditions on LSE

➢ The structure of the matrix for estimating multiple model parameters (估计多个模型参数时的矩阵结构)

$$P(t) = (\Phi(t)^T \Phi(t))^{-1}$$

May be inferred based on the eigenvalues λ_j and the eigenvectors v_j (基于特征值和特征向量得出) $R_\varphi(0) = \sum_{j=1}^{p} \lambda_j v_j v_j^T$

- λ_j reveals the convergence rate (特征值指示收敛率)
- v_j describes the corresponding direction that converges at the rate (特征向量描述收敛率相应的方向)

Summary

- **LSE under Effect of No Model Structure Error** (无模型结构误差对LSE的影响)
- **LSE under Effect of Model Structure Error** (模型结构误差对LSE的影响)
  - ✓E1. *The data set has to be large enough* (数据集要足够大)
  - ✓E2. *The data set has to be rich enough* (数据集要足够充分)
  - ✓E3. *Un-correlate the regressor vector φ(i) and the model error vector e(i)* (不相关的回归向量φ(*i*)和模型误差向量*e*(*i*))
  - ✓E4 . *Maximize the "signal to noise ratio" (SNR)* (最大化信噪比)
- **Converging Rate of the LSE Estimation** (最小二乘估计的收敛率)